ATOMIC STRUCTURE & QUANTUM MECHANICS

Step-by-Step Practice Problems & Solutions Companion

A high-density conceptual resource designed to reinforce the mathematical relationships of atomic theory.

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PART I: CONCEPTUAL PRACTICE PROBLEMS

Concept 1: Subatomic Particles & Net Ionic Ratios

Q1. Calculate the exact number of protons, neutrons, and electrons present in a tripositive ion of iron (Fe^{3+}) with an atomic number of 26 and a mass number of 56.

Q2. An unknown dipositive cation (X^{2+}) has a mass number of 24 and contains an equal number of neutrons and protons. Calculate the total number of electrons in this ion and write the symbol of the parent element.

Concept 2: Nuclear Species Classifications

Q3. Identify the specific relationship (Isotopes, Isobars, Isotones, or Isodiaphers) for each of the following atomic pairs:

(a) ${}^{14}_{6}C$ and ${}^{15}_{7}N$

(b) ${}^{40}_{18}Ar$ and ${}^{40}_{20}Ca$

(c) ${}^{235}_{92}U$ and ${}^{231}_{90}Th$

Q4. Arrange the following chemical species in order of decreasing ionic size and justify your order using nuclear charge concepts: O^{2-}, F^{-}, Na^{+}, Mg^{2+}, and Al^{3+}.

Concept 3: Electromagnetic Radiation & Spectral Gradients

Q5. An FM radio station broadcasts its signal at a frequency of 100 MHz. Calculate the wavelength of this radiation in meters and convert the wavelength to nanometers ($c = 3.0 \times 10^8$ m s^{-1}).

Q6. A laser emits light with a wavelength of 450 nm. Determine the wavenumber ($\bar{\nu}$) of this light in cm^{-1} and calculate the energy of a single photon of this laser light in Joules ($h = 6.626 \times 10^{-34}$ J s).

Concept 4: Einstein's Photoelectric Effect

Q7. The work function of a certain metal surface is 2.4 eV. Determine the threshold frequency (ν_0) of this metal and establish if an incident photon of wavelength 600 nm has enough energy to eject a photoelectron ($h = 6.626 \times 10^{-34}$ J s, 1 eV $= 1.602 \times 10^{-19}$ J).

Q8. A clean metal surface with a threshold wavelength of 310 nm is irradiated with light of wavelength 200 nm. Calculate the maximum kinetic energy (in eV) and the maximum velocity of the ejected photoelectrons ($m_e = 9.11 \times 10^{-31}$ kg).

Concept 5: Bohr's Hydrogenic Orbits (Radii & Energies)

Q9. Calculate the radius of the second Bohr orbit of a He^{+} ion (in pm) and compare it quantitatively with the radius of the third orbit of a Li^{2+} ion.

Q10. The energy of an electron in the ground state of a hydrogen atom is -13.6 eV. Calculate the energy required to excite an electron from the second orbit to the fourth orbit in a hydrogen atom, and compare it to the energy of the second orbit in a Li^{2+} ion.

Concept 6: Rydberg-Ritz Spectral Transitions

Q11. Determine the wavelength of the second line of the Balmer series (transition from $n = 4 \to 2$) in a Hydrogen atom. Given that Rydberg's constant $R_H = 1.097 \times 10^7$ m^{-1}, state the spectral region where this line is observed.

Q12. Calculate the shortest wavelength (series limit) of the Lyman series for a singly ionized helium ion (He^+) and write the answer in terms of Rydberg's constant R_H.

Concept 7: de Broglie Wavelength & Wave-Particle Duality

Q13. Calculate the de Broglie wavelength (in pm) of an electron accelerated through a potential difference of 150 V. (Use the simplified formula $\lambda \approx 1.227/\sqrt{V}$ nm).

Q14. Two particles, an alpha-particle ($m_\alpha \approx 4$ amu) and a proton ($m_p \approx 1$ amu), have identical kinetic energies. Calculate the ratio of their de Broglie wavelengths, $\lambda_p : \lambda_\alpha$.

Concept 8: Heisenberg's Uncertainty Principle

Q15. If the uncertainty in the velocity of a helium atom (mass $= 6.64 \times 10^{-27}$ kg) is measured to be 1.0 m s^{-1}, calculate the minimum uncertainty in its position.

Q16. In a quantum measurement, the uncertainty in position and velocity of a subatomic particle of mass m are found to be numerically equal. Derive the general formula for the minimum velocity uncertainty and calculate its value if $m = 9.11 \times 10^{-31}$ kg.

Concept 9: Quantum Numbers & Orbital Nodes

Q17. Calculate the radial nodes, angular nodes, and total nodes present in:

(a) a $3s$ orbital

(b) a $4d$ orbital

Q18. Write down the complete set of four quantum numbers (n, l, m_l, m_s) for the highest-energy valence electron of a potassium atom ($Z = 19$).

Concept 10: Electronic Configurations & Degenerate Shell Rules

Q19. Write the ground-state electronic configuration of a copper atom ($Z = 29$) and explain why its d-orbital configuration departs from the standard sequential Aufbau filling sequence.

Q20. Determine the total number of unpaired electrons in a cobalt cation in its ground state (Co^{2+}, $Z = 27$) and calculate its spin-only magnetic moment (μ_s) in Bohr Magnetons.

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PART II: DETAILED COMPREHENSIVE SOLUTIONS

Solution to Q1

Parameters given: Iron atom atomic number $Z = 26$, mass number $A = 56$, charge $q = +3$.
Formulas used:

$$\text{Protons } (p) = Z$$
$$\text{Neutrons } (n) = A - Z$$
$$\text{Electrons } (e^-) = Z - q \quad \text{(for a cation)}$$

Step-by-step calculations:

- Protons: $p = 26$.
- Neutrons: $n = 56 - 26 = 30$.
- Electrons: Since it is a Fe^{3+} ion, it has lost 3 electrons. Thus, $e^- = 26 - 3 = 23$.

Answer: Protons = 26, Neutrons = 30, Electrons = 23.

Solution to Q2

Parameters given: Mass number $A = 24$, charge $q = +2$, neutron-proton equality: $N = Z$.
Formulas used: $A = Z + N$. Since $N = Z$, we have $A = 2Z \implies Z = A/2$.
Step-by-step calculations:

- Atomic Number: $Z = 24/2 = 12$.
- The element with atomic number 12 is Magnesium (Mg).
- For a dipositive cation (Mg^{2+}), the number of electrons is: $e^- = Z - 2 = 12 - 2 = 10$.

Answer: Electrons = 10; Element = Magnesium (Mg).

Solution to Q3 **Parameters given:** Pairs to evaluate.

Formulas & Concepts used:

- (a) $^{14}_{6}\mathrm{C}$ (neutrons $14-6=8$) and $^{15}_{7}\mathrm{N}$ (neutrons $15-7=8$) are **Isotones**.
- (b) $^{40}_{18}\mathrm{Ar}$ (mass 40) and $^{40}_{20}\mathrm{Ca}$ (mass 40) are **Isobars**.
- (c) $^{235}_{92}\mathrm{U}$ (protons 92, neutrons 143) and $^{231}_{90}\mathrm{Th}$ (protons 90, neutrons 141) both share the same neutron-proton excess: $(N-Z)=143-92=51$ (for U), and $(N-Z)=141-90=51$ (for Th). Thus they are **Isodiaphers**.

Answer: (a) Isotones, (b) Isobars, (c) Isodiaphers.

Solution to Q4 **Parameters given:** Species: O^{2-}, F^-, Na^+, Mg^{2+}, Al^{3+}.

Formulas & Concepts used:

- These are **isoelectronic species** because they all contain exactly 10 electrons.
- For isoelectronic species, the ionic radius is inversely proportional to the nuclear charge (Z).

Step-by-step calculations:

$$\begin{aligned}
\mathrm{O}^{2-}&: Z=8 \implies \text{Ratio } Z/e^- = 0.8\\
\mathrm{F}^{-}&: Z=9 \implies \text{Ratio } Z/e^- = 0.9\\
\mathrm{Na}^{+}&: Z=11 \implies \text{Ratio } Z/e^- = 1.1\\
\mathrm{Mg}^{2+}&: Z=12 \implies \text{Ratio } Z/e^- = 1.2\\
\mathrm{Al}^{3+}&: Z=13 \implies \text{Ratio } Z/e^- = 1.3
\end{aligned}$$

A higher nuclear charge exerts a stronger electrostatic pull on the same 10 electrons, pulling the electron cloud closer and decreasing the radius. **Answer:** $O^{2-} > F^- > Na^+ > Mg^{2+} > Al^{3+}$.

Solution to Q5 **Parameters given:** Frequency $\nu = 100$ MHz $= 1.0\times10^8$ Hz, speed of light $c = 3.0\times10^8$ m s^{-1}.

Formulas used: $c=\nu\lambda \implies \lambda = \frac{c}{\nu}$.
Step-by-step calculations:

- Wavelength in meters:
$$\lambda = \frac{3.0\times10^8 \text{ m s}^{-1}}{1.0\times10^8 \text{ s}^{-1}} = 3.0 \text{ m}$$
- Conversion to nanometers: Since 1 m $=10^9$ nm,
$$\lambda = 3.0\times10^9 \text{ nm}$$

Answer: Wavelength $=3.0$ m (or 3.0×10^9 nm).

Solution to Q6 **Parameters given:** Wavelength $\lambda = 450$ nm $=4.5\times10^{-7}$ m.

Formulas used:

$$\bar{\nu} = \frac{1}{\lambda}$$
$$E = \frac{hc}{\lambda}$$

Step-by-step calculations:

- Wavenumber:
$$\bar{\nu} = \frac{1}{4.5\times10^{-7}\text{ m}} = 2.22\times10^6 \text{ m}^{-1} = 2.22\times10^4 \text{ cm}^{-1}$$

- Energy:

$$E = \frac{(6.626 \times 10^{-34} \text{ J s}) \times (3.0 \times 10^{8} \text{ m s}^{-1})}{4.5 \times 10^{-7} \text{ m}} = 4.417 \times 10^{-19} \text{ J}$$

Answer: Wavenumber $= 2.22 \times 10^{4}$ cm^{-1}; Photon Energy $= 4.42 \times 10^{-19}$ J.

Solution to Q7

Parameters given: Work function $W_0 = 2.4 \text{ eV} = 2.4 \times (1.602 \times 10^{-19} \text{ J}) = 3.845 \times 10^{-19}$ J, incident photon wavelength $\lambda = 600 \text{ nm} = 6.0 \times 10^{-7}$ m.

Formulas used:

$$W_0 = h\nu_0 \implies \nu_0 = \frac{W_0}{h}$$

$$E_{\text{photon}} = \frac{hc}{\lambda}$$

Step-by-step calculations:

- Threshold frequency:

$$\nu_0 = \frac{3.845 \times 10^{-19} \text{ J}}{6.626 \times 10^{-34} \text{ J s}} = 5.80 \times 10^{14} \text{ Hz}$$

- Energy of incident photon:

$$E_{\text{photon}} = \frac{(6.626 \times 10^{-34} \text{ J s}) \times (3.0 \times 10^{8} \text{ m s}^{-1})}{6.0 \times 10^{-7} \text{ m}} = 3.313 \times 10^{-19} \text{ J}$$

- Comparison: Convert E_{photon} to eV:

$$E_{\text{photon}} = \frac{3.313 \times 10^{-19} \text{ J}}{1.602 \times 10^{-19} \text{ J/eV}} \approx 2.07 \text{ eV}$$

  Since $E_{\text{photon}} = 2.07$ eV is strictly less than the work function $W_0 = 2.4$ eV, no photoelectric ejection can occur.

Answer: $\nu_0 = 5.80 \times 10^{14}$ Hz; Ejection = No (insufficient energy).

Solution to Q8 **Parameters given:** Threshold wavelength $\lambda_0 = 310$ nm, incident wavelength $\lambda = 200$ nm.
Formulas used:

$$K.E._{\max} = E_{\text{photon}} - W_0 = hc\left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right) = \frac{hc}{\lambda} - \frac{hc}{\lambda_0}$$
$$K.E._{\max} = \frac{1}{2}m_e v^2 \implies v = \sqrt{\frac{2 \cdot K.E._{\max}}{m_e}}$$

Step-by-step calculations:

- Convert hc to eV nm: $hc \approx 1240$ eV nm.
- Energy of incident photon: $E_{\text{photon}} = \frac{1240 \text{ eV nm}}{200 \text{ nm}} = 6.2$ eV.
- Work function: $W_0 = \frac{1240 \text{ eV nm}}{310 \text{ nm}} = 4.0$ eV.
- Kinetic energy: $K.E._{\max} = 6.2 \text{ eV} - 4.0 \text{ eV} = 2.2$ eV.
- Conversion to Joules: $2.2 \text{ eV} \times (1.602 \times 10^{-19} \text{ J/eV}) = 3.524 \times 10^{-19}$ J.
- Velocity:
$$v = \sqrt{\frac{2 \times 3.524 \times 10^{-19} \text{ J}}{9.11 \times 10^{-31} \text{ kg}}} \approx 8.80 \times 10^5 \text{ m s}^{-1}$$

Answer: Kinetic Energy = 2.2 eV; Velocity = 8.80×10^5 m s^{-1}.

Solution to Q9 **Parameters given:** Orbit $n_1 = 2$ for He^+ ion ($Z = 2$), orbit $n_2 = 3$ for Li^{2+} ion ($Z = 3$).
Formulas used: $r_n = 52.9\frac{n^2}{Z}$ pm.
Step-by-step calculations:

- Radius of He^+ ($n = 2$):
$$r_{\text{He}^+,n=2} = 52.9 \times \frac{2^2}{2} = 52.9 \times 2 = 105.8 \text{ pm}$$
- Radius of Li^{2+} ($n = 3$):
$$r_{\text{Li}^{2+},n=3} = 52.9 \times \frac{3^2}{3} = 52.9 \times 3 = 158.7 \text{ pm}$$
- Quantitative Ratio:
$$\frac{r_{\text{He}^+,n=2}}{r_{\text{Li}^{2+},n=3}} = \frac{105.8}{158.7} = \frac{2}{3} \quad (2:3 \text{ ratio})$$

Answer: $r(\text{He}^+) = 105.8$ pm, $r(\text{Li}^{2+}) = 158.7$ pm. Ratio is 2 : 3.

Solution to Q10 **Parameters given:** Hydrogen ground state energy $E_1 = -13.6$ eV, transition $n = 2 \to 4$ in H.
Formulas used:

$$E_n = \frac{-13.6}{n^2} \text{ eV}$$
$$\Delta E = E_4 - E_2 = -13.6\left(\frac{1}{4^2} - \frac{1}{2^2}\right) \text{ eV} = 13.6\left(\frac{1}{4} - \frac{1}{16}\right) \text{ eV}$$

Step-by-step calculations:

- Energy level 2 in H: $E_2 = -13.6/4 = -3.4$ eV.
- Energy level 4 in H: $E_4 = -13.6/16 = -0.85$ eV.

- Energy difference: $\Delta E = -0.85 - (-3.4) = 2.55$ eV.
- Second orbit energy of Li^{2+} ion ($Z = 3$):

$$E_{\mathrm{Li}^{2+},n=2} = -13.6 \times \frac{3^2}{2^2} = -13.6 \times \frac{9}{4} = -30.6 \text{ eV}$$

Answer: Excitation energy $\Delta E = 2.55$ eV; Li^{2+} energy $= -30.6$ eV.

Solution to Q11

Parameters given: Transition $n = 4 \to 2$, $R_H = 1.097 \times 10^7\ \mathrm{m}^{-1}$.
Formulas used:

$$\bar{\nu} = \frac{1}{\lambda} = R_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

Step-by-step calculations:

- Calculate the transition term:

$$\frac{1}{\lambda} = R_H \left(\frac{1}{2^2} - \frac{1}{4^2} \right) = R_H \left(\frac{1}{4} - \frac{1}{16} \right) = \frac{3}{16} R_H$$

- Compute wavelength:

$$\lambda = \frac{16}{3R_H} = \frac{16}{3 \times 1.097 \times 10^7\ \mathrm{m}^{-1}} \approx 4.862 \times 10^{-7}\ \mathrm{m} = 486.2\ \mathrm{nm}$$

- This line lies in the **visible light** region (Balmer series).

Answer: Wavelength = 486.2 nm; region = Visible light (cyan-blue spectral line).

Solution to Q12

Parameters given: Series limit of Lyman series for He^{+} ion ($Z = 2$).
Formulas used:

$$\frac{1}{\lambda} = R_H Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

Step-by-step calculations:

- For the Lyman series, the terminal orbit is $n_1 = 1$.
- For the shortest wavelength (series limit), the electron falls from infinity: $n_2 = \infty$.
- Inserting $Z = 2$ and $n_2 = \infty$:

$$\frac{1}{\lambda} = R_H (2)^2 \left(\frac{1}{1^2} - \frac{1}{\infty^2} \right) = 4R_H$$

- Thus:

$$\lambda = \frac{1}{4R_H}$$

Answer: Shortest wavelength $\lambda = \frac{1}{4R_H}$.

Solution to Q13 **Parameters given:** Potential $V = 150$ V.
Formulas used: $\lambda = \frac{1.227}{\sqrt{V}}$ nm.
Step-by-step calculations:

- Substitute $V = 150$:
$$\lambda \approx \frac{1.227}{\sqrt{150}} = \frac{1.227}{12.247} \text{ nm} \approx 0.1002 \text{ nm}$$
- Convert to picometers: Since 1 nm = 1000 pm,
$$\lambda \approx 100.2 \text{ pm}$$

Answer: Wavelength $\lambda = 100.2$ pm.

Solution to Q14 **Parameters given:** Equal kinetic energy for proton and alpha particle ($K.E._p = K.E._\alpha$). Mass ratio: $m_\alpha = 4m_p$.
Formulas used: $\lambda = \frac{h}{\sqrt{2m \cdot K.E.}} \implies \lambda \propto \frac{1}{\sqrt{m}}$ (since $K.E.$ is constant).
Step-by-step calculations:

- Set up the ratio equation:
$$\frac{\lambda_p}{\lambda_\alpha} = \sqrt{\frac{m_\alpha}{m_p}}$$
- Substitute the mass relationship:
$$\frac{\lambda_p}{\lambda_\alpha} = \sqrt{\frac{4m_p}{m_p}} = \sqrt{4} = 2$$

Answer: Ratio $\lambda_p : \lambda_\alpha = 2 : 1$.

Solution to Q15 **Parameters given:** Velocity uncertainty $\Delta v = 1.0$ m s^{-1}, mass $m = 6.64 \times 10^{-27}$ kg, $h = 6.626 \times 10^{-34}$ J s.
Formulas used:
$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi} \implies \Delta x \cdot (m\Delta v) \geq \frac{h}{4\pi} \implies \Delta x \geq \frac{h}{4\pi m \Delta v}$$
Step-by-step calculations:
$$\Delta x \geq \frac{6.626 \times 10^{-34} \text{ J s}}{4 \times 3.14159 \times (6.64 \times 10^{-27} \text{ kg}) \times (1.0 \text{ m s}^{-1})} = \frac{6.626 \times 10^{-34}}{8.344 \times 10^{-26}} \approx 7.94 \times 10^{-9} \text{ m}$$
Answer: Minimum uncertainty in position $\Delta x = 7.94 \times 10^{-9}$ m (or 7.94 nm).

Solution to Q16 **Parameters given:** Numerical equality: $\Delta x = \Delta v$. Mass $m = 9.11 \times 10^{-31}$ kg, $h = 6.626 \times 10^{-34}$ J s.
Formulas used:
$$\Delta x \cdot (m\Delta v) \geq \frac{h}{4\pi}$$
$$\text{Since } \Delta x = \Delta v \implies m(\Delta v)^2 \geq \frac{h}{4\pi} \implies \Delta v_{\min} = \sqrt{\frac{h}{4\pi m}}$$
Step-by-step calculations:
$$\Delta v_{\min} = \sqrt{\frac{6.626 \times 10^{-34} \text{ J s}}{4 \times 3.14159 \times (9.11 \times 10^{-31} \text{ kg})}} = \sqrt{\frac{6.626 \times 10^{-34}}{1.1448 \times 10^{-29}}} \approx \sqrt{5.788 \times 10^{-5}} \approx 7.61 \times 10^{-3} \text{ m s}^{-1}$$

Answer: General formula: $\Delta v_{\min} = \sqrt{\frac{h}{4\pi m}}$; value for electron $= 7.61 \times 10^{-3}$ m s^{-1}.

Solution to Q17 **Formulas used:**

$$\text{Radial Nodes} = n - l - 1$$
$$\text{Angular Nodes} = l$$
$$\text{Total Nodes} = n - 1$$

Step-by-step calculations:

- (a) For $3s$ orbital: $n = 3, l = 0$.
  - Radial nodes: $3 - 0 - 1 = 2$.
  - Angular nodes: $l = 0$.
  - Total nodes: $3 - 1 = 2$.
- (b) For $4d$ orbital: $n = 4, l = 2$.
  - Radial nodes: $4 - 2 - 1 = 1$.
  - Angular nodes: $l = 2$.
  - Total nodes: $4 - 1 = 3$.

Answer: (a) $3s$: Radial = 2, Angular = 0, Total = 2. (b) $4d$: Radial = 1, Angular = 2, Total = 3.

Solution to Q18 **Parameters given:** Potassium atom atomic number $Z = 19$, configuration: $[Ar]4s^1$ (valence electron in $4s$).

Formulas & Rules used: Standard quantum numbers limits: n represents the outer energy shell, l represents the orbital shape, m_l orbital orientation, m_s spin state.

Step-by-step assignments:

- For $4s$ subshell: $n = 4$.
- Since it is an s-orbital: $l = 0$.
- Since $l = 0$: m_l must be 0.
- For a single electron in an orbital: spin $m_s = +1/2$ (or $-1/2$).

Answer: $n = 4, l = 0, m_l = 0, m_s = +1/2$.

Solution to Q19 **Parameters given:** Chromium atomic number $Z = 24$.

Step-by-step analysis:

- Standard filling order (Aufbau) would predict: $[Ar]3d^4 4s^2$.
- However, the actual ground state configuration is: $[Ar]3d^5 4s^1$.
- **Thermodynamic Reason 1: Exchange Energy ($E_{\mathbf{ex}}$):** Exchange energy is stabilizing and is proportional to the number of parallel spin pairs that can exchange positions within degenerate orbitals:

$$P_{\text{ex}} = \frac{k(k-1)}{2} \text{ where } k \text{ is the number of electrons of parallel spin}$$

  For a $3d^4 4s^2$ system, the d-subshell has 4 parallel spins: Exchanges $= \frac{4(3)}{2} = 6$. For a $3d^5 4s^1$ system, the d-subshell has 5 parallel spins: Exchanges $= \frac{5(4)}{2} = 10$. This represents an extra 4 exchanges, providing highly stabilizing exchange energy.

- **Thermodynamic Reason 2: Charge Distribution Symmetry:** A half-filled subshell ($3d^5$) has a highly symmetrical, spherically symmetric distribution of charge, which minimizes inter-electron repulsion.

Answer: Configuration: $[Ar]3d^5 4s^1$. Stabilized by high exchange energy (10 exchanges) and symmetrical charge shielding.

Solution to Q20 **Parameters given:** Cobalt atomic number $Z = 27$, ion Co^{2+}.

Formulas used:

$$\text{Spin-only magnetic moment } (\mu_s) = \sqrt{N(N+2)} \text{ B.M.} \quad \text{(where } N = \text{number of unpaired electrons)}$$

Step-by-step calculations:

- Ground state of neutral Cobalt atom ($Z = 27$): $[Ar]3d^7 4s^2$.
- Cobalt cation (Co^{2+}) loses two valence electrons from the outer $4s$ shell first: $[Ar]3d^7$.
- Orbital filling of $3d^7$ subshell (via Hund's rule):
  - Orbital 1: $\uparrow\downarrow$ (paired)
  - Orbital 2: $\uparrow\downarrow$ (paired)
  - Orbital 3: $\uparrow$ (unpaired)
  - Orbital 4: $\uparrow$ (unpaired)
  - Orbital 5: $\uparrow$ (unpaired)

  This yields exactly $N = 3$ unpaired electrons.
- Calculate the magnetic moment:

$$\mu_s = \sqrt{3(3+2)} = \sqrt{15} \approx 3.87 \text{ B.M.}$$

Answer: Unpaired electrons = 3; Spin-only magnetic moment = 3.87 B.M.